Chapter 9 — Acid-Base Chemistry

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Acid-base concepts and ionic product of water

Q.1 Compare Arrhenius, Bronsted-Lowry and Lewis acid-base concepts and explain the ionic product of water.

Arrhenius acids produce H^+ in water and bases produce OH^-. Bronsted-Lowry acids donate protons and bases accept protons. Lewis acids accept an electron pair and Lewis bases donate an electron pair. Conjugate acid-base pairs differ by one proton.

Water self-ionizes: $H_2O \rightleftharpoons H^+ + OH^-$. The ionic product is $K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C. In pure water, $[H^+] = [OH^-] = 1.0 \times 10^{-7}$ mol dm^{-3}, so pH = 7.

Important Conceptual Short Questions — Topic 1

1. Why does pure water have pH 7 at 25 °C?

Self-ionization gives equal concentrations of H^+ and OH^-, each 1.0×10^{-7} mol dm^{-3}. Therefore pH = $-\log(10^{-7}) = 7$.

2. How does the Lewis concept broaden the Bronsted-Lowry concept?

Lewis theory includes reactions without proton transfer. Any electron-pair acceptor is an acid and any electron-pair donor is a base, so species such as BF_3 can act as acids.

3. Why does K_w increase with temperature?

Self-ionization of water is endothermic. Raising temperature favours ionization, increasing both $[H^+]$ and $[OH^-]$ and therefore K_w.

4. How are conjugate acid-base pairs related?

They differ by one proton. When an acid donates H^+, it becomes its conjugate base; when a base accepts H^+, it becomes its conjugate acid.

Topic 2 — pH, pOH and acid ionization constant

Q.2 Explain pH, pOH, Ka and calculations for strong and weak acids.

$pH = -\log[H^+]$ and $pOH = -\log[OH^-]$. At 25 °C, pH + pOH = 14. Strong acids ionize completely, so $[H^+]$ can be found directly from concentration. Weak acids ionize partially and are described by $K_a = [H^+][A^-]/[HA]$.

For a weak acid of concentration c and small ionization, $[H^+] \approx \sqrt{(K_a c)}$. A larger K_a means a stronger acid. $pK_a = -\log K_a$, so a smaller pK_a means a stronger acid.

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Important Conceptual Short Questions — Topic 2

1. How does dilution affect the pH of a strong acid?

Dilution lowers $[H^+]$, so pH increases toward 7. It does not become basic because the solution remains acidic.

2. Why does a weak acid have a higher pH than a strong acid of the same concentration?

A weak acid ionizes only partially, so it produces fewer H^+ ions than a strong acid at the same concentration.

3. How does Ka indicate acid strength?

A larger Ka means the equilibrium lies further toward ionized products, giving more H^+ ions and a stronger acid.

4. Why is pKa often more convenient than Ka?

Ka values are very small and cover many powers of ten. pKa converts them into a compact logarithmic scale where lower values mean stronger acids.

Topic 3 — Common ion effect and buffers

Q.3 Explain the common ion effect and how buffer solutions resist pH change.

The common ion effect is the suppression of ionization of a weak electrolyte by adding a strong electrolyte with a common ion. For example, adding sodium acetate to acetic acid increases $[CH_3COO^-]$ and shifts the acid ionization equilibrium backward.

A buffer resists pH change when small amounts of acid or base are added. An acidic buffer contains a weak acid and its salt, such as CH_3COOH/CH_3COONa. A basic buffer contains a weak base and its salt, such as NH_3/NH_4Cl. The Henderson-Hasselbalch equation for an acidic buffer is $pH = pK_a + \log([salt]/[acid])$.

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Important Conceptual Short Questions — Topic 3

1. Why does adding sodium acetate reduce the ionization of acetic acid?

The added acetate ion is a common ion. By Le Chatelier’s principle, the equilibrium shifts backward, reducing H^+ production.

2. How does an acidic buffer neutralize added acid?

The conjugate base from the salt combines with added H^+ to form the weak acid, so the increase in $[H^+]$ is small.

3. How does an acidic buffer neutralize added base?

The weak acid donates H^+ to neutralize added OH^-, forming water and the conjugate base, so pH changes only slightly.

4. Why is buffer capacity limited?

A buffer can neutralize only a limited amount of added acid or base. Once one component is used up, the pH changes sharply.

Topic 4 — Solubility product and salt hydrolysis

Q.4 Explain solubility product, its applications and salt hydrolysis.

For a sparingly soluble salt A_xB_y, $K_{sp} = [A^{n+}]^x[B^{m-}]^y$ at equilibrium. If the ionic product is less than K_{sp}, the solution is unsaturated; if equal, saturated; if greater, precipitation occurs. The common ion effect decreases solubility.

Salt hydrolysis is the reaction of salt ions with water. Salts of strong acids and strong bases are neutral. Salts of strong acids and weak bases are acidic. Salts of weak acids and strong bases are basic. Salts of weak acids and weak bases depend on the relative strengths of the ions.

Important Conceptual Short Questions — Topic 4

1. How can Ksp predict whether a precipitate forms?

Calculate the ionic product from the mixed ion concentrations. If it exceeds Ksp, the solution is supersaturated and precipitation occurs until equilibrium is restored.

2. Why does the common ion effect decrease the solubility of a sparingly soluble salt?

Adding a common ion shifts the dissolution equilibrium backward, so less solid dissolves.

3. Why is an ammonium chloride solution acidic?

NH_4^+ is the conjugate acid of a weak base and reacts with water to produce H_3O^+. Cl^- does not hydrolyse, so the solution becomes acidic.

4. Why is a sodium acetate solution basic?

CH_3COO^- is the conjugate base of a weak acid and reacts with water to produce OH^-. Na^+ does not hydrolyse, so the solution becomes basic.

Topic 5 — Indicators and titration curves

Q.5 Explain acid-base indicators and how titration curves are used to select an indicator.

An indicator is a weak organic acid or base whose unionized and ionized forms have different colours. The end point is the colour change, while the equivalence point is the point at which stoichiometric neutralization is complete. A suitable indicator changes colour near the equivalence point.

In a strong acid-strong base titration, the pH change near equivalence is steep and phenolphthalein or methyl orange may be used. In a weak acid-strong base titration, the equivalence point is basic, so phenolphthalein is suitable. In a strong

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acid-weak base titration, the equivalence point is acidic, so methyl orange is suitable.

Important Conceptual Short Questions — Topic 5

1. Why is phenolphthalein suitable for a weak acid-strong base titration?

The equivalence point is above 7 because the conjugate base hydrolyses. Phenolphthalein changes colour in the basic range, close to the steep part of the curve.

2. How does an indicator show the end point of a titration?

As pH changes near the equivalence point, the indicator shifts between its acid and base forms, producing a visible colour change.

3. Why are end point and equivalence point not always identical?

The equivalence point is a stoichiometric point, while the end point is the observed indicator colour change. A good indicator makes them very close.

4. Why is no sharp indicator suitable for a weak acid-weak base titration?

The pH change near equivalence is too gradual, so no indicator gives a sharp, reliable colour change.

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